Amendment to OSF Registration TDW6C

Registration: 10.17605/OSF.IO/TDW6C - *Energy performance measurement models in continuous process industries under ISO 50001/50006* (Protocol v2, April 2026)
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Methodological specification: Validation Protocol C-2.3

1. Purpose and timing

The registered protocol specifies a single human reviewer and does not describe AI-assisted charting. This amendment prospectively records the revised review scope, corpus definition, eligibility interpretation, search-supplementation rule, and AI-assisted charting and quality-control design to be used for the full corpus.

The amendment takes effect before full-corpus production extraction. Earlier development and operational pilots were used only to test and refine the workflow and do not contribute evidence to the final validation estimates. One operational pilot identified loss of unsaved reviewer-form data; the affected decisions were excluded from blind-validation claims, and the corrected workflow was requalified before production.

The single-reviewer basis is unchanged. Bernardo Ribeiro remains the sole reviewer and sole adjudicator of record. The two large-language-model families are charting tools, not independent reviewers, and agreement between them is used only to route human review effort.

2. Amendments to the registered review methods

2.1 Search supplementation and final corpus closure

Backward and forward citation searching will be conducted on the included corpus and reported in accordance with PRISMA-S. A pre-specified cause-routed supplementation rule will be applied before the final review corpus is frozen.

Let S be the set of records included through citation searching, R the subset retrievable by the registered Scopus/Web of Science strategy, and M = S \ R. Supplementation is triggered when S > 0 and either |M|/|S| > 0.10 or |M| >= 5. Misses will be classified by cause: search-string failure, engineering-database indexing gap, thesis/dissertation coverage, or grey-literature coverage. The corresponding remedy will be, respectively, expansion and re-running of the registered strings, a supplementary Compendex search, a dedicated dissertation source or documented practical exclusion, and a targeted grey-literature source or declared limitation.

All additional records will be deduplicated and screened through the same registered flow. Citation searching and the retrievability assessment will be repeated on newly included records until a complete pass adds no new included records. Only then will the final review corpus be frozen. If S = 0, the null citation-search yield will be reported as uninformative about database recall. Residual recall limitations remain, including doctoral theses excluded by database document-type filters, relevant calibration terminology absent from the final registered string, and grey literature not reached by the registered databases.

2.2 Scope of claims and research questions

The review maps what is reported in the searched, retrieved and eligible literature. It does not apply an a priori adequacy matrix and does not test operational effectiveness at the Sines refinery. The research questions are interpreted as follows:

1. SQ1 maps the distribution of EnPI/EnB models by sector, organisation type, normative context, year and model family in the eligible review corpus.
2. SQ2 maps operational-variability mechanisms and model limitations reported by detected analytical-corpus studies.
3. SQ3 maps adaptations proposed for continuous-process contexts, without claiming that those adaptations are effective.
4. SQ4 maps applications, requirements and barriers reported as potentially transferable to petroleum refining. Operational validation at Sines is a separate empirical component of the thesis.

2.3 Analytical corpus (Corpus B)

Corpus-B membership will be determined by production mode against a pre-specified sector list. In addition to the five registered sectors - petroleum refining, petrochemicals, cement, pulp and paper, and steel - the main list is prospectively expanded to continuous chemical production, industrial gases, and glass produced by continuous melting.

Membership will not be defined by the non-stationarity mechanisms or model limitations being studied, because those are extracted outcomes. Continuous-flow cases outside the pre-specified list will be reported separately as

exploratory cases unless a dated corpus decision expands the main list before their results are counted. Results will be reported by sector, allowing the original five-sector and refining-only views to remain identifiable. Petrochemical operations will not be treated as equivalent to petroleum refineries for SQ4 transferability claims.

2.4 Eligibility interpretation

The registered full-text threshold is retained: a record is eligible if it presents, proposes, evaluates, or critiques a relevant model with enough detail to identify and characterise that model. Methodological critiques, conceptual proposals, and unvalidated models may therefore be eligible. The presence, type, and quality of validation are charted variables, not eligibility requirements.

An earlier development implementation had inadvertently converted this threshold into a stricter conjunction. That implementation was corrected before full-corpus production extraction. No production record was extracted or retained under the stricter threshold, so no sensitivity analysis is required.

3. AI-assisted charting and human validation

All available full texts will be processed independently by two large-language-model families using the same frozen data-item schema and versioned prompts. Records meeting the eligibility criterion will proceed to data charting. Raw outputs, prompt versions, displayed model labels, application versions, run dates, and adjudication records will be retained.

Outputs will be compared field by field after deterministic formatting and label normalisation. Model agreement is not treated as evidence of correctness or as inter-rater reliability. Human review will be allocated according to the methodological consequence of each field:

1. **Consequential disagreements.** Every disagreement concerning methodological eligibility, Corpus-B classification, paper type, or borderline review/synthesis eligibility will be adjudicated by the reviewer blind-first: the reviewer decides from the source before model values are revealed. The blind decision, final decision, evidence, decision rule, and any post-reveal change will be retained.
2. **Corpus-B false-negative control.** A candidate union will be generated from model flags, a deterministic terminology search across all full texts, and a recall-oriented counter-anchored model pass. Every candidate will receive a human decision. Confirmed false negatives will expand the terminology list and trigger a new pass; this loop closes only when a complete pass produces no new candidates. The terminology search will also be applied to all title/abstract exclusions, together with a simple random sample of 50 exclusions not identified by that search (seed 44; take all if fewer than 50 remain). This procedure reduces but does not eliminate the risk of papers invisible to all detectors.
3. **Corpus-B analytical fields.** The human treatment is determined after the reviewer locks the initial-wave Corpus-B roster. If |B| <= 25, all Corpus-B papers are charted blind-first. If 25 < |B| <= 60, a sector-priority head of 25 papers plus a simple random subsample n_sub = min(15, |B| - 25) of the remainder is charted blind-first. Consequently, all papers are blind-first through |B| = 40; papers beyond 40 receive AI prefill followed by field-by-field human verification. Subsample results are reported as x/n_sub with the exact one-sided 95% upper confidence bound where relevant. If |B| > 60, Level-B work pauses until a dated amendment note records the revised scope and supervisor discussion.
4. **Bibliometric and descriptive fields.** Divergent Level-A cells are adjudicated with both model outputs visible. Concordant Level-A cells are evaluated through a simple random sample of 40 eligible papers (seed 44), re-extracted blind. For each field, the report gives the observed error count e_v/n_v and an exact Clopper-Pearson 95% confidence interval. Consequential cells subject to a census are excluded from the sampled denominator for that field.
5. **Traceability and consistency checks.** Every AI-extracted value must carry source evidence that can be verified against the full text; failures are routed to human review. Reviewer self-consistency is measured after at least two weeks by re-deciding 30 consequential decisions blind to the originals and model outputs: 20 sampled from borderline-tagged decisions and 10 from the remainder (seed 44, without replacement; take all borderline decisions and fill to 30 if fewer than 20 exist). Run-to-run model instability is assessed separately for each model by reprocessing a simple random sample of 30 papers (seed 44) from the proceeding corpus at the end of the run under the same recorded configuration. Any consequential-field flip receives human review; a changed application version or displayed model label makes the comparison configuration-non-comparable rather than a fixed-configuration canary.

For positive or present findings, human evidence requires a supporting quotation and physical page. For an absence-like value, the reviewer may instead record that the item was not reported after full-text review, together with an explicit absence or applicability rationale; an invented quotation or page is not permitted. When a record is adjudicated ineligible, Level-A fields are not applicable and remain null. Any unresolved disagreement on an eligible record requires human adjudication before final consolidation.

4. Pre-specified escalation and workload rules

The following thresholds are operational control rules, not statistical acceptance tests:

- If e_v >= 3 errors are found for any Level-A field in the audit, that field is rechecked across the corpus and a root-cause note is recorded.
- Any confirmed Corpus-B false negative expands the terminology list and restarts the candidate search; three or more confirmed false negatives trigger supervisor discussion of a corpus-wide sector re-screen.
- More than one flip in the 30-decision reviewer test-retest triggers re-adjudication of all borderline-tagged decisions.
- A run-to-run canary flip rate above 2% is reported as a model-instability limitation; above 5%, all flipped fields are re-reconciled.
- Any grounding failure is routed to human review.

If available human time becomes insufficient, the Level-A audit may be reduced from 40 to 30 papers. The resulting loss of precision will be stated explicitly: with zero observed errors, the one-sided 95% upper bound changes from approximately 7.2% to 9.5%. No model value will acquire automatic precedence merely because time is exhausted. If a further reduction is necessary, work will pause and a dated deviation will identify any non-core descriptive outputs omitted from the synthesis. The false-negative census, blind-first adjudication of consequential disagreements, and the Corpus-B treatment prescribed by the active size branch will not be reduced.

Execution is ordered so that analytical-corpus membership is settled before its deep charting: dual model processing and reconciliation; blind-first adjudication of membership-critical disagreements and completion of the false-negative candidate census; lock of the initial-wave Corpus-B roster; selection of the Corpus-B size branch; Corpus-B charting; remaining adjudication; Level-A audit; test-retest and instability canary. The initial-wave roster remains provisional until citation-searching and supplementation closure.

5. Reporting, limitations, and reproducibility

Validation results will be reported separately by field and instrument: model-divergence rates, blind reveal-change counts, Level-A audited error counts with exact confidence intervals, false-negative-census outcomes, movements into or out of Corpus B after human review, grounding-check results, reviewer test-retest flips, and model canary flip rates. No pooled composite error rate, Cohen's kappa, or model-agreement coefficient will be used. The reviewer is the sole adjudicator, not an infallible reference standard; test-retest measures self-consistency rather than accuracy.

The design provides exhaustive human review only for explicitly enumerable sets, including candidate-union members and specified consequential disagreements. Concordant eligible cells not covered by a census or sample are provisionally accepted. Residual risk remains for records outside the candidate union, title/abstract exclusions outside the audit, unsampled Level-A cells, and unavailable full texts. Claims will therefore be limited to the coverage actually achieved and will not describe Corpus-B membership or field accuracy as completely closed.

At the initial database-retrieval freeze, 459 of 596 deduplicated records were available in full text and 137 (23.0%) were not retrieved. Retrieval differed by prior screening decision and DOI status: 85.0% of records marked Included were retrieved compared with 65.4% marked Maybe; 91.6% of records with a DOI were retrieved compared with 30.3% without one. Validation claims apply only to the retrieved corpus. Missingness will be described by screening decision, DOI status, year, document type, and sector signals, and all counts will be updated after final corpus closure.

The use of one human reviewer is a declared limitation relative to JBI guidance recommending independent review in source selection and piloted or verified data charting. PRISMA-ScR will be used for reporting, with PRISMA-S for the search and PRISMA-trAIce as a supplementary transparency checklist for AI involvement.

The AI systems are accessed through consumer applications that do not provide guaranteed dated model snapshots, controllable sampling parameters, or request identifiers. Exact re-execution against future application versions is therefore not expected to reproduce the same outputs. Reproducibility rests on the deposited methodological protocol, frozen schema and prompts, displayed model and application identifiers, raw model outputs, human decisions and evidence, and final validation logs.